

FACULTY OF SCIENCE, AGRICULTURE & ENGINEERING**DEPARTMENT OF ENGINEERING**

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ASSESSMENT of GAs No.	GA 1 and GA2
STUDENT SUBMISSION	INDIVIDUAL ☒ GROUP ☐
ASSIGNED DURATION	10 HOURS
DATE SUBMITTED	XX / XX / 2026

GAs STUDENT SUBMISSION COVER PAGE**Signed:**

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*The guideline or best practice for disclosure is as per **Appendix A** below.

Appendix A

1. AI Tool Use Disclosure

For each AI tool that you have used, provide a brief, high-level summary:

1.	Tool Name and Version:	DeepSeek
	Purpose:	finding solution for errors ladder logic diagram was giving me
	Scope of Use	

2.	Tool Name and Version:	ChatGBT
	Purpose:	For finding right papers base on my topic
	Scope of Use	

**Add as many tables as you require or delete tables as appropriate.*

2. AI Tool Use Detailed Disclosure by Category

For every instance where you used an AI tool, you must provide a detailed explanation.

A. AI as a Writing Tool (*e.g., drafting, summarising, paraphrasing, editing, proofreading, image generation*)

Requirement	Explanation
Specific use	5 Mathematical and Analytical Considerations
Input/Prompt	prompt
Validation process	
Required evidence	

**Add as many tables as you require.*

B. AI as an Idea Generation Tool (*e.g., brainstorming, outlining, topic selection, structural design*)

Requirement	Explanation
Specific use	Summarizing 6 Literature and Engineering Knowledge, and the English I used
Input/Prompt	input
Validation process	
Required evidence	

**Add as many tables as you require.*

C. AI as a Code Generation or Development Tool (*e.g., generating code snippets, debugging, explaining code, writing test cases, any other technical/implementation aspects in your design project*)

Requirement	Explanation
Specific use	generating ladder diagram snippets, helping me to improve ladder logic diagrams, and helping me to use CODESYS and WPL Software
Input/Prompt	prompt
Validation process	
Required evidence	

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1 Introduction

Traffic congestion is a common problem in many cities, especially on busy roads like the N1 in Johannesburg. Many vehicles use these roads every day, and intersections often become crowded. One serious problem is that emergency vehicles such as ambulances, police cars, and fire trucks are delayed at traffic lights. These delays can increase response time and may lead to dangerous situations [1][8]. Modern traffic systems are now moving towards automation and smart control. Research shows that using sensors, a timer, and an intelligent control system can improve traffic flow and reduce delay [3][6]. One important feature is emergency vehicle pre-emption, in which traffic lights automatically change to give priority to emergency vehicles [4].

In this project, a Programmable Logic Controller (PLC) is used to design a smart traffic light system. The system is based on ladder logic and includes normal traffic operation, pedestrian control, emergency override, and night mode flashing. The design is simple, reliable, and suitable for real-world use, especially in developing regions.

2 Problem Identification and System Description

The main problem is designing a traffic light control system that manages traffic efficiently while prioritizing emergency vehicles. The system must also ensure safety for pedestrians and handle faults properly.

Table 1: PLC Input and Outputs

Inputs	PLC Addresses	Description
NS-Loop	X0	Detects vehicles in North/South direction
EW-Loop	X1	Detects vehicles in the East/West direction
Ped-Request	X2	Pedestrian push button
Emergency	X3	Detects an emergency vehicle
Night Mode	X4	Activates night mode
Fault	X5	Detect system fault
Reset	X6	Reset the system
outputs		
NS-Signal	Y0	Controls North/South traffic lights
EW-Signal	Y1	Controls East /West traffic light
Ped-Signal	Y2	Control pedestrian signals
Emergency Relay	Y3	Controls emergency override

Table 2 PLC Memory and Timers

Memory	PLC Addresses	Timers	PLC Addresses
NS-Green	M0	T-NS-Green	T0
NS-Yellow	M1	T-NS-Yellow	T1
EW-Green	M2	T-EW-Green	T2
EW-Yellow	M3	T-EW-Yellow	T3
Ped walks	M4	T-Ped-walk	T4
Emergency Mode	M5	T-Blink	T5
Blink	M6		
Normal Traffic Enable	M7		

The main objective of the system is to control traffic safely and efficiently. It must reduce waiting times, allow smooth traffic flow, and give immediate priority to emergency vehicles. According to research, combining sensor data with control logic improves intersections and safety [3].

3. System Operation and Ladder Logic Analysis

The system operation is based on your ladder logic design. It works in different modes depending on the inputs.

3.1 Normal Traffic Operation

In normal mode, the system follows a fixed sequence. The North/South direction starts with a green light, then changes to yellow, and then the East/West direction gets a green light. The sequence continues in a loop. Timers are used to control how long each light stays on. The green light is set to 30 seconds, and the yellow light is set to 5 seconds. These values are commonly used in traffic systems to balance safety and efficiency [2]. The pedestrian crossing is included in the sequence. When a pedestrian presses the button, the system allows crossing after the current traffic phase ends. The pedestrian signal stays on for 10 seconds, which is enough time for safe crossing [6].

3.2 Emergency Mode Operation

Emergency mode is activated when the input X3 Emergency is ON. In the ladder Logic, this is handled in rung 1. Once activated, the system enters Emergency mode and stays there until it is reset. In this mode, all normal traffic signals are turned off, and the emergency relay is activated. This allows the emergency vehicle to pass through the intersection without delay. Research shows that emergency signal pre-emption is very important and must override all other operations immediately [4].

3.3 Night Mode Operation

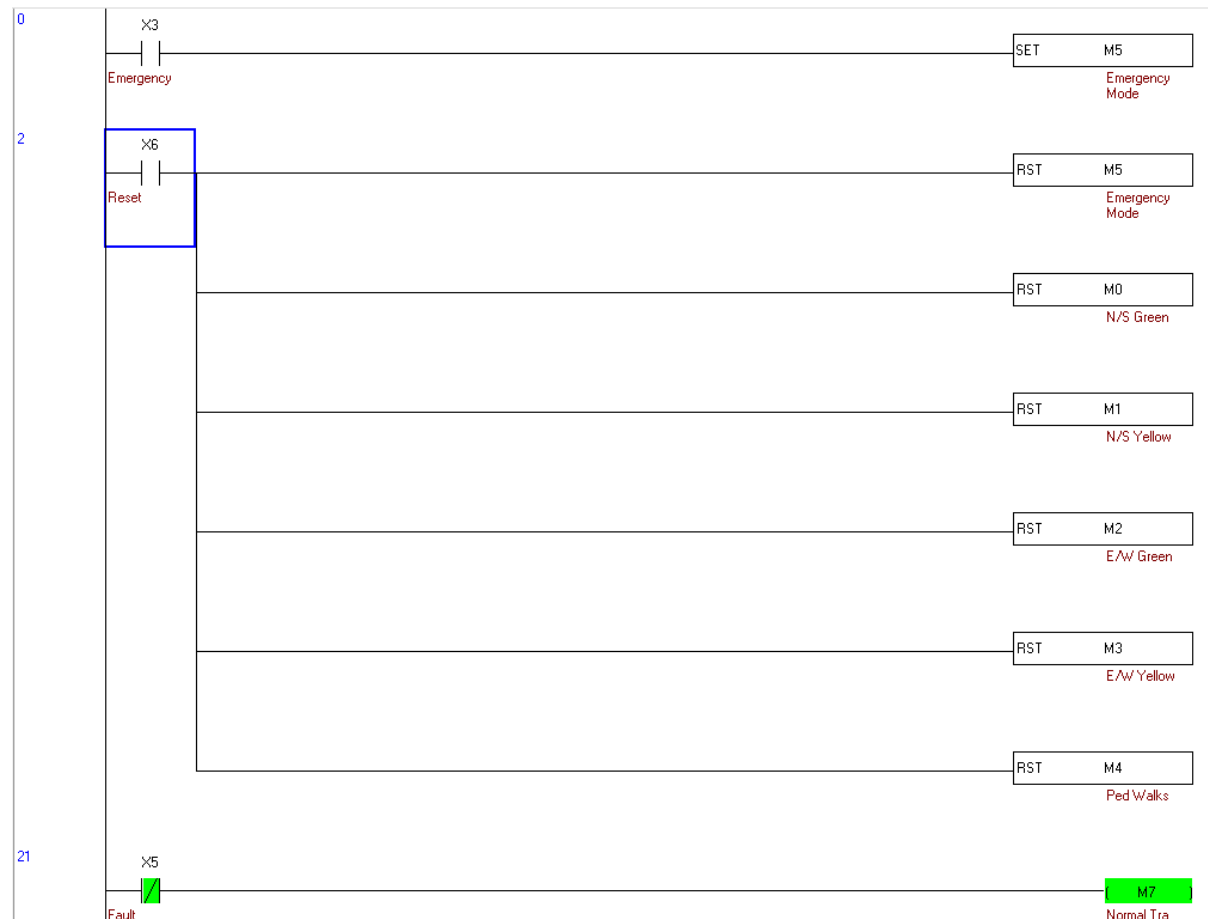
Night mode is activated using the X4 Night Mode input. In this mode, the traffic lights do not follow the normal sequence. Instead, they flash using a timer. A blink timer of 5s is used to

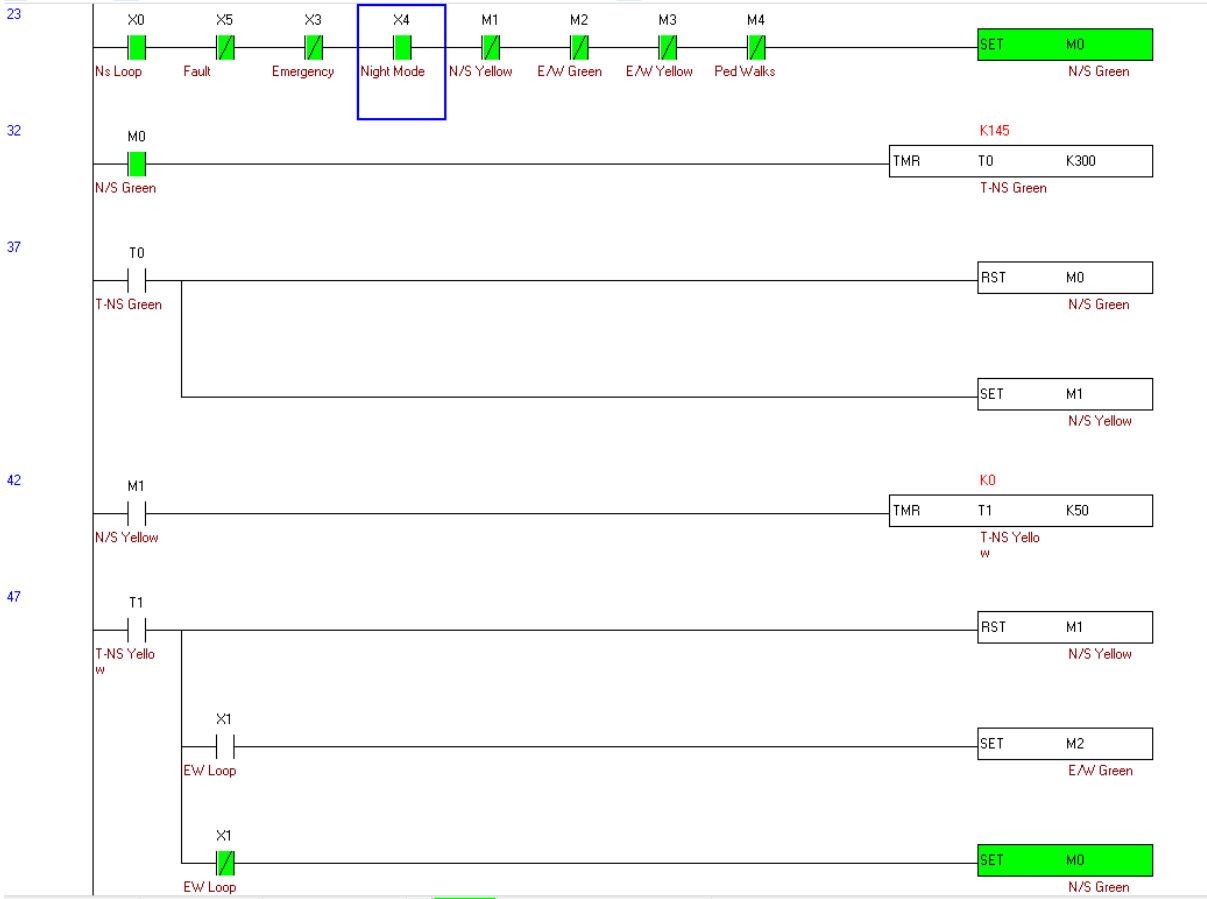
turn the lights ON and OFF repeatedly. This reduces energy usage and is suitable when traffic is low at night. Studies show that flashing signals are effective in low traffic conditions and help reduce unnecessary waiting [5].

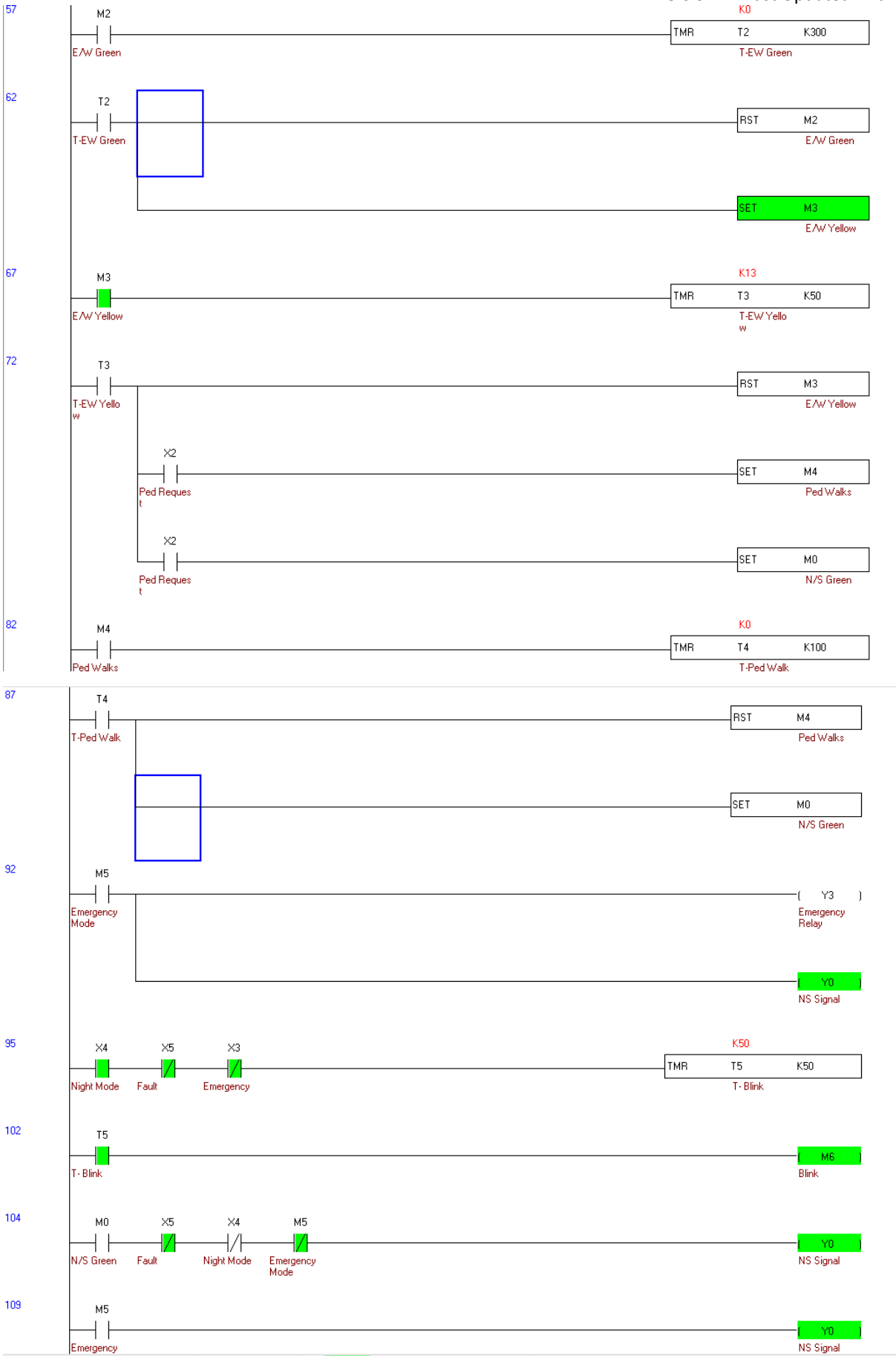
3.4 Fault handling

The system includes a fault input X5. When a fault occurs, the system enters a safe state. This prevents incorrect operation and ensures safety. Fault handling is important in industrial systems because it protects both the equipment and the users. A reset signal is required to return the system to normal.

The figure below shows the Ladder Logic Diagram using WPL Software







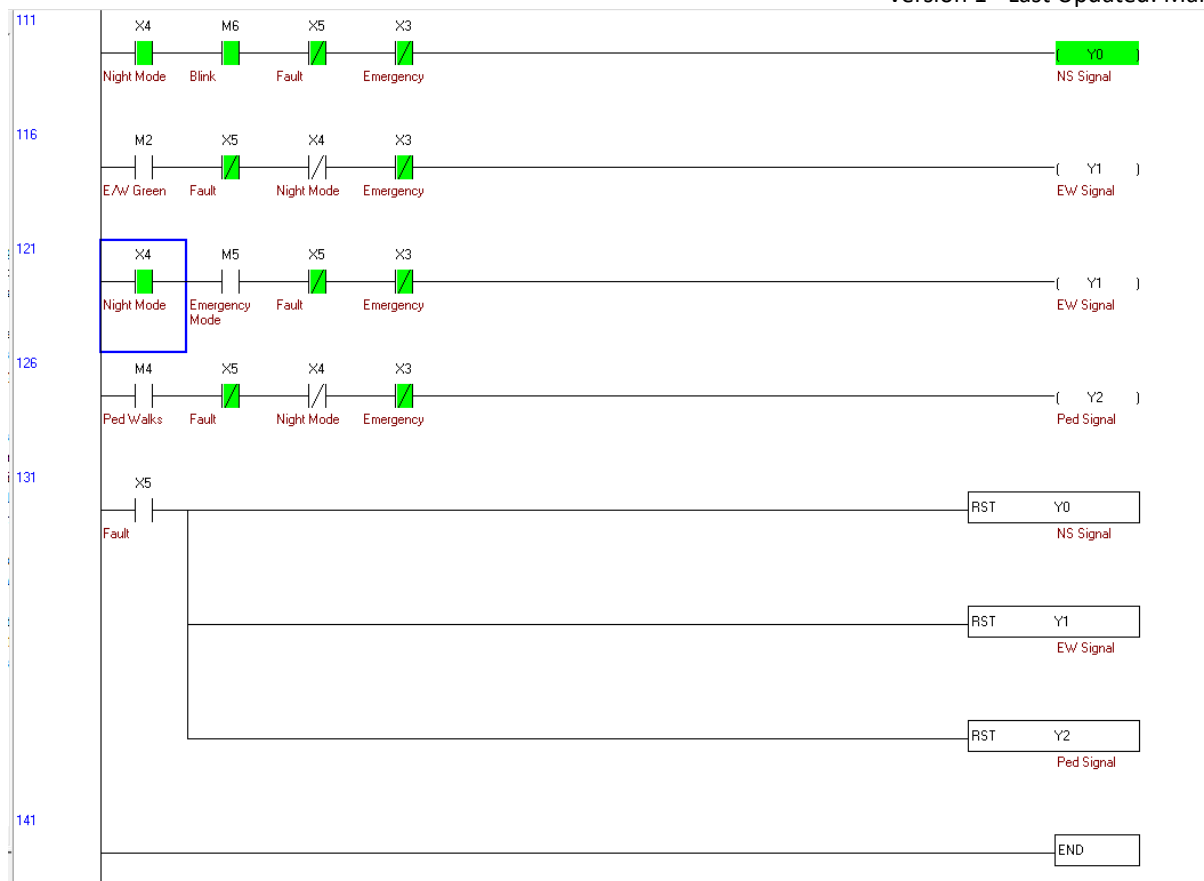


Figure 1 Ladder logic Diagram for a traffic light

4. Application of Engineering Principles

The system applies basic engineering principles from electrical and control engineering. Timers are used to control the duration of each traffic light. Logical operations such as AND and OR are used to make decisions in the ladder logic. Memory elements such as Set and Reset are used to store the current state of the system.

The design also uses a sensor to detect vehicles and pedestrians. These sensors provide real-time input to the PLC, which allows the system to respond correctly. According to research, sensor-based systems improve traffic efficiency and reduce delay [3],[7].

5 Mathematical and Analytical Considerations

Mathematica is used in setting the timer values and controlling the sequence of operations. The green time, the yellow time, and pedestrian time are chosen based on standard traffic engineering practices. The blink timer in night mode is set to 5S, which creates a visible flashing effect. Proper timing is important because it affects both safety and traffic flow. Research shows that correct timing reduces congestion and improves system performance [2].

6 Literature and Engineering Knowledge.

This project is supported by several research studies. These studies show that modern traffic systems use sensors, communication systems, and automation to improve performance.

In industrial automation, a PLC is commonly used because it is strong, reliable, and easy to program. The ladder logic used in this project follows standard PLC design methods, where inputs are read, processed using logic, and then outputs are controlled. The use of memory bits such as M0 NS_Green and M5 Emergency Mode shows how the PLC system stores and manages different states during operation.

Modern traffic systems depend heavily on sensors and automation. Studies show that vehicle detection sensors, such as loop detectors, help the system understand traffic demand in real time [3]. This idea is directly used in this project through the inputs X0 NS_Loop and I1 EW_Loop, which allow the system to decide when to change traffic lights. Research also explains that emergency vehicle detection is an important part of smart traffic systems. Some systems use GPS or wireless communication to detect emergency vehicles and send signals to traffic lights [7]. In this project, this concept is simplified using the input X3, which activates emergency mode. When this input is ON, the ladder logic immediately overrides all normal operations, which follows the same principle [4].

Another important concept from research is traffic signal timing. Studies show that fixed-time control systems are simple and effective for many intersections, especially where traffic patterns are predictable [2]. This project uses fixed timers for green, yellow, and pedestrian phases. Even though more advanced systems use adaptive timing, fixed timing is still widely used because it is easy to implement and reliable. The use of pedestrian control in this project is also supported by research. Studies show that pedestrian push-button systems improve safety and reduce unnecessary delays [6]. In the ladder logic, the pedestrian request is only activated when needed, which makes the system more efficient.

Night mode operation is another feature. Research shows that flashing signals are suitable during low traffic conditions because they reduce waiting time and energy usage [5]. In this project, the blinking timer of 5s is used to create a flashing effect, which follows this principle. In addition, studies on traffic systems show that emergency pre-emption can significantly reduce delays for emergency vehicles and improve response time [8],[9]. This supports the design choice of giving the highest priority to the emergency mode in the ladder logic.

Overall, this project combines basic PLC programming with ideas from modern research. Even though the system is simple, it follows the same principles used in advanced traffic control systems. This makes the design practical, cost-effective, and suitable for real-world use.

7. Evaluation and Discussions

The system was evaluated based on its ability to meet the design objectives. The results show that the system works as expected. Emergency vehicles are given immediate priority, which reduces delays. Normal traffic flows in a controlled sequence, and pedestrians are allowed to cross safely. Night mode reduces unnecessary operation when traffic is low. However, the system has some limitations. It uses fixed timing and does not adapt to changing traffic conditions. Also, it depends on the sensors working correctly.

Despite these limitations, the system is reliable and cost-effective. It is suitable for real-world use, especially in areas where advanced systems are too expensive.

8. Conclusion

This project designed a PLC-based traffic light control system with emergency vehicle bypass. The system will improve traffic flow, increase safety, and reduce delays for emergency vehicles. The use of ladder logic makes the system simple and easy to understand. The design also follows real engineering principles and is supported by research. Overall, the system is practical, efficient, and suitable for implementation in developing regions.

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